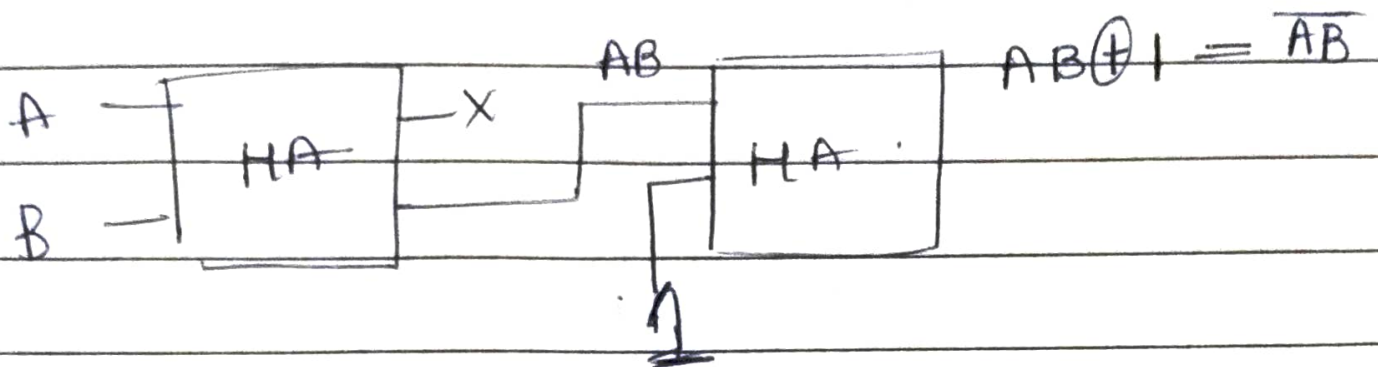
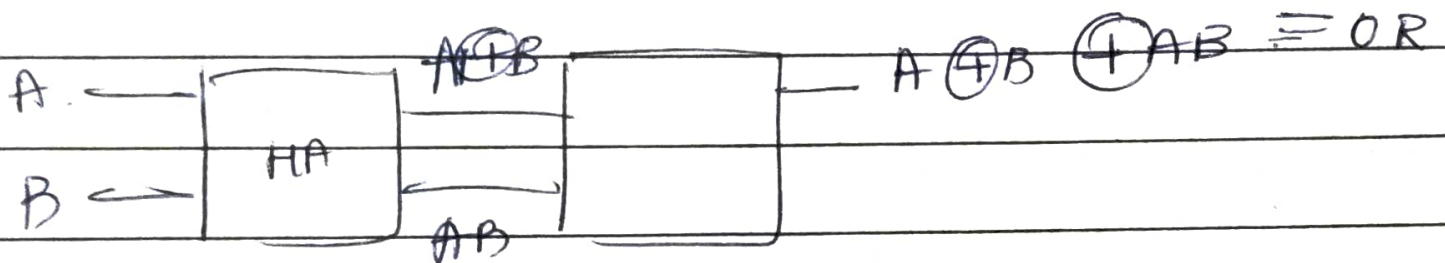


HA as NAND Gate



HA as OR Gate



$$A \oplus B \oplus AB$$

Full Adders

$$Sum = \sum m(1, 2, 4, 7)$$

$$Carry = \sum m(3, 5, 6, 7)$$

4-bit Carry Look Ahead Adder

$A+B$ generates carry only if $A=B=1$

$A+B$ propagates carry either if one of A or $B=1$

~~Diff = 2m~~

$Y = \text{sel ? } 1'b0 : 1'b1 \rightarrow \text{Converts into mux}$
 or

if (sign)

$Y = a;$

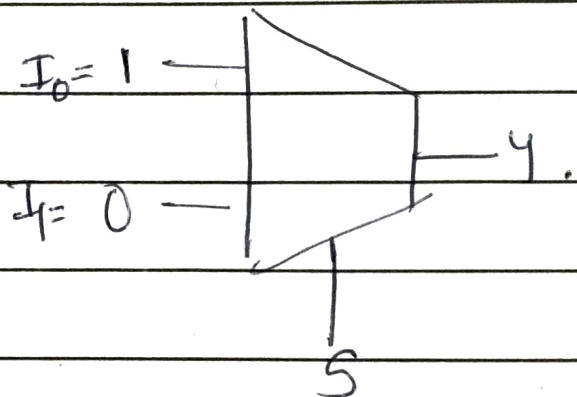
else $Y = b;$

NOT Gate ~~or~~ Using MUX.

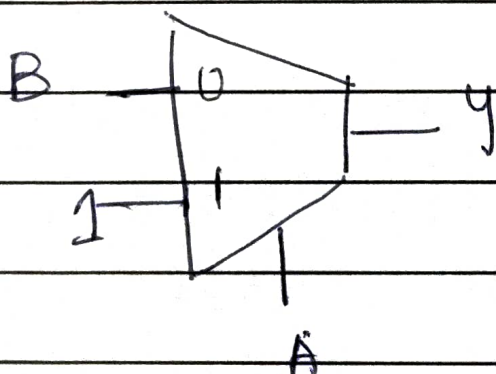
$$Y = \bar{S}I_0 + SI_1$$

$$= \bar{S}(1) + S(0)$$

$$Y = \bar{S}$$



OR Gate Using MUX



A	B	Y	Y
0	0	0	B
0	1	1	B
1	0	1	1
1	1	1	1

$$\bar{A}B + A$$

$$(A+B) (\bar{A}+B)$$